

DevOps Lab Programs

Program-6: Deploy an automated build pipeline using Docker hub

Structure of the program:

Program6/

→ Dockerfile

→ package.json

→ package-lock.json

→ node_modules

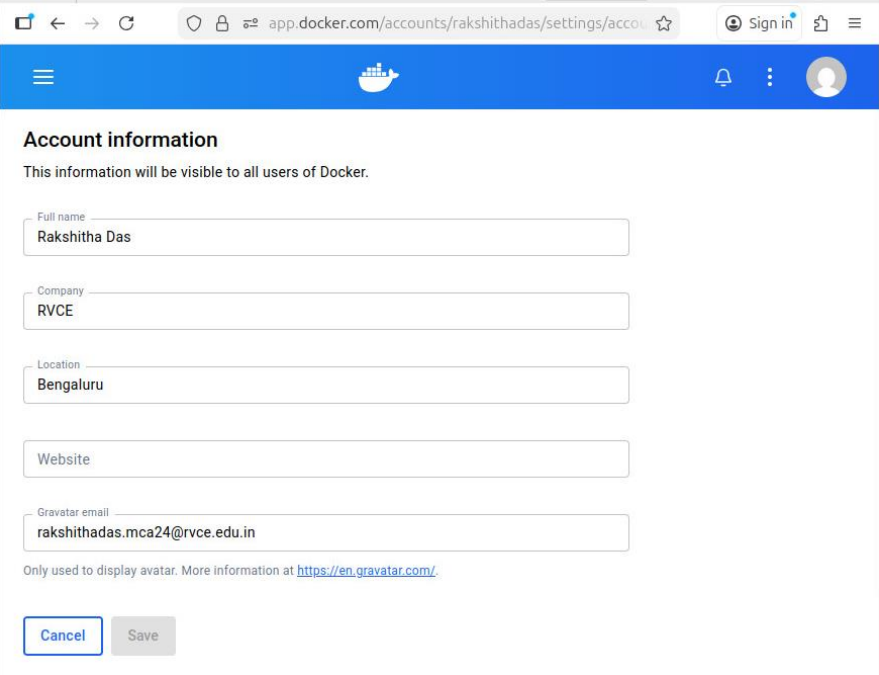
Steps for execution:

Step 1: Create the Personal Access Token(PAT) in github

- Create a repo(program6)
- Navigate to the settings → Developers settings → Generate tokens
- Generate the token with specific scope:

Write: packages

Read: packages



The screenshot shows the Docker account settings page in a web browser. The URL is `app.docker.com/accounts/rakshithadas/settings/account`. The page has a blue header with the Docker logo and a user profile icon. The main content area is titled "Account information" and includes a warning: "This information will be visible to all users of Docker." Below this, there are several input fields: "Full name" (Rakshitha Das), "Company" (RVCE), "Location" (Bengaluru), "Website" (empty), and "Gravatar email" (rakshithadas.mca24@rvce.edu.in). A note at the bottom of the Gravatar email field states: "Only used to display avatar. More information at <https://en.gravatar.com/>." At the bottom left, there are "Cancel" and "Save" buttons.

Step 2: Create folder program6 and write the appropriate files required

Step 3: Create a Docker file and build the images

```
1rv24mc083-rakshi
GNU nano 8.1
FROM jenkins/jenkins:lts

USER root

RUN apt-get update && apt-get install -y docker.io

USER jenkins
```

Step 3: Tag the image for docker hub and login to the docker hub using PAT

```
1rv24mc083-rakshithadas@1rv24mc083rakshithadas: ~/Docker/lab6
=> [1/5] FROM docker.io/library/python:3.11-slim@sha256:193fdd0bbcb3d2ae612bd6cc354 0.0s
=> [internal] load build context 0.1s
=> => transferring context: 39.03kB 0.0s
=> CACHED [2/5] WORKDIR /app 0.0s
=> CACHED [3/5] COPY requirements.txt . 0.0s
=> CACHED [4/5] RUN pip install --no-cache-dir -r requirements.txt 0.0s
=> [5/5] COPY . . 0.2s
=> exporting to image 0.1s
=> => exporting layers 0.1s
=> => writing image sha256:5f211a0185ab1d296dfcb364c3725612be74f83f70b8e348de31e231 0.0s
=> => naming to docker.io/library/lab6 0.0s
1rv24mc083-rakshithadas@1rv24mc083rakshithadas:~/Docker/lab6$ docker tag lab6 rakshithadas/
lab6:latest
1rv24mc083-rakshithadas@1rv24mc083rakshithadas:~/Docker/lab6$ docker login -u rakshithadas

Info → A Personal Access Token (PAT) can be used instead.
To create a PAT, visit https://app.docker.com/settings

Password:

WARNING! Your credentials are stored unencrypted in '/home/1rv24mc083-rakshithadas/.docker/
config.json'.
Configure a credential helper to remove this warning. See
https://docs.docker.com/go/credential-store/

Login Succeeded
1rv24mc083-rakshithadas@1rv24mc083rakshithadas:~/Docker/lab6$ docker push rakshithadas/lab6
:latest
The push refers to repository [docker.io/rakshithadas/lab6]
1f38f301e9f3: Pushed
ab5015581f7a: Mounted from rakshithadas/prog6
0f0801ab4830: Mounted from rakshithadas/prog6
f730f6bab040: Mounted from rakshithadas/prog6
720ee7aae3ad: Mounted from rakshithadas/prog6
655ff69eb9c8: Mounted from rakshithadas/prog6
5c988eaa8862: Mounted from rakshithadas/prog6
70a290c5e58b: Mounted from rakshithadas/prog6
latest: digest: sha256:9615103296c5742a0d18cddb5a89b483d05130079bc443f97a2dd453fbe9e166 siz
e: 1993
1rv24mc083-rakshithadas@1rv24mc083rakshithadas:~/Docker/lab6$ docker tag lab6 ghcr.io/raksh
ithadas/lab6:latest
1rv24mc083-rakshithadas@1rv24mc083rakshithadas:~/Docker/lab6$ export GITHUB_PAT=ghp_HLJ6YSz
HCD.../7Zee3DQe10A4bYv72B6v...
```

Step 5: Push the image to the docker hub and github

```
1rv24mc083-rakshithadas@1rv24mc083rakshithadas: ~/Docker/lab6
create mode 100644 Docker/lab6/node_modules/type-is/HISTORY.md
create mode 100644 Docker/lab6/node_modules/type-is/LICENSE
create mode 100644 Docker/lab6/node_modules/type-is/README.md
create mode 100644 Docker/lab6/node_modules/type-is/index.js
create mode 100644 Docker/lab6/node_modules/unpipe/HISTORY.md
create mode 100644 Docker/lab6/node_modules/unpipe/LICENSE
create mode 100644 Docker/lab6/node_modules/unpipe/README.md
create mode 100644 Docker/lab6/node_modules/unpipe/index.js
create mode 100644 Docker/lab6/node_modules/unpipe/package.json
create mode 100644 Docker/lab6/node_modules/vary/HISTORY.md
create mode 100644 Docker/lab6/node_modules/vary/LICENSE
create mode 100644 Docker/lab6/node_modules/vary/README.md
create mode 100644 Docker/lab6/node_modules/vary/index.js
create mode 100644 Docker/lab6/node_modules/vary/package.json
create mode 100644 Docker/lab6/node_modules/wrappy/LICENSE
create mode 100644 Docker/lab6/node_modules/wrappy/README.md
create mode 100644 Docker/lab6/node_modules/wrappy/package.json
create mode 100644 Docker/lab6/node_modules/wrappy/wrappy.js
create mode 100644 Docker/lab6/package-lock.json
create mode 100644 Docker/lab6/package.json
create mode 100644 Docker/lab6/requirements.txt
1rv24mc083-rakshithadas@1rv24mc083rakshithadas: ~/Docker/lab6$ git branch -M main
1rv24mc083-rakshithadas@1rv24mc083rakshithadas: ~/Docker/lab6$ git push -u origin main
Username for 'https://github.com': RakshithaDas
Password for 'https://RakshithaDas@github.com':
Enumerating objects: 14876, done.
Counting objects: 100% (14876/14876), done.
Delta compression using up to 8 threads
Compressing objects: 100% (10274/10274), done.
Writing objects: 100% (14876/14876), 440.28 MiB | 723.00 KiB/s, done.
Total 14876 (delta 4205), reused 14819 (delta 4175), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (4205/4205), done.
remote: warning: See https://gh.io/lfs for more information.
remote: warning: File snap/firefox/common/.mozilla/firefox/azd73ycc.default/storage/default
/https+++web.whatsapp.com/idb/2181795718meogdaerlo-ts.sqlite is 75.06 MB; this is larger th
an GitHub's recommended maximum file size of 50.00 MB
remote: warning: GH001: Large files detected. You may want to try Git Large File Storage -
https://git-lfs.github.com.
To https://github.com/RakshithaDas/lab6.git
* [new branch]      main -> main
branch 'main' set up to track 'origin/main'.
1rv24mc083-rakshithadas@1rv24mc083rakshithadas: ~/Docker/lab6$
```

Step 6: Add the local project to the github:

- Add the remote repo, if not already linked

git remote add origin <https://github.com/rakshithadas/lab6.git>

- If the remote repo already exists(but incorrect)

Git remote set-url origin <https://github.com/rakshithadas/lab6.git>

Step 7: Check the uploads in both registries

In the gitHub:

- Check whether the source code is present in the repository
- Under your profile in the packages the particular image is present

The screenshot shows a web browser window displaying the GitHub repository page for 'lab6' by user 'RakshithaDas'. The browser's address bar shows the URL 'github.com/RakshithaDas/lab6'. The repository page includes a header with the user's profile, repository name, and navigation tabs like 'Code', 'Issues', 'Pull requests', 'Actions', 'Projects', 'Wiki', 'Security', and 'Insights'. Below the header, there are buttons for 'Pin', 'Watch', 'Fork', and 'Star'. The main content area shows a list of files and folders, including '.cache', '.config', '.docker', '.local', '.npm', '.pki/nssdb', 'Docker', 'Downloads', 'Pictures/Screenshots', 'examPractice', 'prog6', 'secondfile', 'secondfile1', 'snap', 'tec7', and '.bash_history'. Each item shows the number of commits and the time since the last commit. On the right side, there is an 'About' section with a description, 'Activity' section with stars, watching, and forks counts, 'Releases' section with a link to 'Create a new release', and 'Packages' section with a link to 'Publish your first package'.

File/Folder	Commit Count	Last Commit Time
.cache	2 commit	last week
.config	2 commit	last week
.docker	2 commit	last week
.local	2 commit	last week
.npm	2 commit	last week
.pki/nssdb	2 commit	last week
Docker	initial commit	18 minutes ago
Downloads	2 commit	last week
Pictures/Screenshots	2 commit	last week
examPractice	2 commit	last week
prog6	initial commit	4 hours ago
secondfile	2 commit	last week
secondfile1	2 commit	last week
snap	2 commit	last week
tec7	2 commit	last week
.bash_history	2 commit	last week

In the dockerHub:

Under your repository check whether the particular mage rakshithadas/lab6 is visible

The screenshot shows the Docker Hub interface for the repository `docker/rakshithadas/lab6`. The page is viewed from the user's perspective, showing the repository details and settings.

Repository Details:

- Repository:** `rakshithadas/lab6`
- Last pushed:** less than a minute ago
- Repository size:** 48.8 MB
- Stars:** 0
- Downloads:** 0

Actions:

- [Add a description](#)
- [Add a category](#)

Docker commands:

```
docker push rakshithadas/lab6:tagname
```

Tags:

Tag	OS	Type	Pulled	Pushed
latest	Image	less than 1 day	1 minute	

[See all](#)

Repository overview INCOMPLETE

An overview describes what your image does and how to run it. It displays in [the public view of your repository](#) once you have pushed some content.

[Add overview](#)

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hub.docker.com/repository/registry-1.docker.io/rakshithadas/lab6/settings#image-insights-settings-card